

Workshop 1 – Kaggle

Mechanisms of Action (MoA) Prediction – Kaggle Competition

Course: Systems Analysis

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1 Introduction

Predicting the mechanisms of action (MoA) of chemical compounds is a key challenge in bioinformatics and drug discovery, as it allows us to identify how compounds affect cells before conducting costly laboratory experiments. The MoA Prediction competition addresses this challenge through a multi-label machine learning approach, in which participants must build models capable of estimating, for each sample, the probability of activation of one or more of the 206 available MoAs.

The dataset provided combines gene expression information, cell viability, and experimental metadata, creating a highly dimensional and complex scenario. The models must handle nonlinear relationships and random processes present in both the underlying biology and the prediction algorithms, ensuring reliable and calibrated results in an environment where variables are highly interconnected and class imbalance is extreme. This competition represents a real-world case study where data science, systems engineering, and molecular biology converge. It allows us to explore how advanced preprocessing, modeling, and probabilistic evaluation techniques can be integrated to solve complex biological prediction problems, ensuring that the system outputs are interpretable, reproducible, and useful for decision-making in biomedical research.

2 Competition summary

The MoA Prediction competition focuses on developing machine learning models capable of predicting the mechanisms of action (MoA) of chemical compounds from high-dimensional biological data, with the goal of accelerating research in bioinformatics and drug discovery. Participants were provided with gene expression data, cell viability data, and experimental metadata (treatment type, dose, and duration), and were required to predict one or more of the 206 official MoAs for each sample.

The dataset combines input variables (g-variables, c-variables, and metadata) with multi-label target variables, where each compound may have multiple biological effects. The problem is approached as a multi-label classification task, with an emphasis on calibrated probabilistic predictions, evaluated using log-loss, which penalizes both errors and overconfident incorrect predictions.

The system faces hard constraints, such as the 206 MoA limit, the exclusive use of available variables, data normalization and clipping, and not considering control samples as positives. It also must handle soft constraints, including high dimensionality, extreme class imbalance, computational cost, and the need for reproducibility in the face of random initializations or validation splits.

Finally, operational rules are applied to ensure consistency and quality: each sample must provide probabilities for all classes, respect the data scale, and maintain the distribution of rare classes during validation. These conditions align the models with the competition’s objectives and standards, ensuring reliable and consistent predictions.

3 Systems analysis

3.1 System components

The MoA Prediction competition can be conceptualized as a system composed of multiple elements that work together to generate predictions of mechanisms of action. Inputs include:

- High-dimensional biological data, represented by more than 800 gene expression variables (g-variables).
- Cell characteristics, with approximately 100 variables (c-variables) that reflect the response of each cell.
- Experimental metadata:
 - Type of compound (`cp_type`).
 - Duration of treatment (`cp_time`).
 - Dose (`cp_dose`).

These entries are processed through different stages that make up the system processes, notably:

- Data preprocessing, including normalization, dimensionality reduction, and selection of relevant features.
- Machine learning modeling, with neural networks, gradient boosting, and ensemble models, designed to capture complex, nonlinear relationships between variables.

As outputs, the system delivers predictions of action mechanisms, represented as probability vectors associated with the 207 officially evaluated mechanisms. The evaluation is performed using metrics such as log loss, which reflect both the accuracy and confidence of the predictions, providing useful feedback for adjusting the preprocessing, feature selection, and model hyperparameters.

3.2 Relationships between elements

The MoA Prediction competition can be considered a complex system in which elements do not interact in isolation, but rather through dynamic relationships that generate critical dependencies and potentially nonlinear

effects. Among the inputs, g-variables (gene expression levels) and c-variables (cellular characteristics) show significant correlations in many cases. This occurs because the activation of certain genes triggers observable cellular responses, such that a variation in gene expression can be directly reflected in changes in cell viability, proliferation, or morphology. In addition, genes often act in interconnected networks or metabolic pathways, generating cascading effects that manifest simultaneously in different variables. Added to this are shared experimental conditions, since the application of a treatment, its exposure time, or its dose simultaneously affect both gene expression and cellular characteristics, introducing structural correlations. In this context, experimental metadata act as key modulators of the relationships between inputs:

- `cp_type` distinguishes between control and treated samples, establishing a fundamental difference: in the former, the activation of mechanisms of action is not expected, while in the latter it may occur depending on other conditions.
- `cp_time` introduces a temporal factor, given that gene expression and cell viability evolve as the duration of treatment increases.
- `cp_dose` regulates the magnitude of the effect, modulating the intensity of the responses observed in both genes and cellular characteristics.

Data preprocessing plays a key role in this framework, as it not only normalizes and reduces the dimensionality of the information, but also mitigates the impact of excessive correlations and outliers. In this way, it establishes a balance between the variables before training the model. Feature selection, meanwhile, reinforces the dependency between inputs and processes, as it identifies which genes or cellular characteristics are most relevant based on the type, timing, and dose of treatment, which directly shapes the structure of the model.

During training, machine learning models integrate all these interactions. A change in an input, such as a difference in `cp_type`, can amplify or attenuate the prediction of certain mechanisms of action, while the model learns complex, nonlinear relationships between g-variables and c-variables modulated by metadata. Each prediction vector thus depends on the quality of the inputs and the model's ability to capture these interactions, which reflect the inherent sensitivity of the system.

Evaluation using metrics such as log loss not only measures the model’s accuracy but also the confidence of its predictions. This makes evaluation a critical component that feeds back into decisions about normalization, variable selection, and hyperparameter tuning. In this sense, some inputs are more sensitive than others:

- `cp_type` has a direct impact on differentiating between control and treated samples, which significantly affects the output.
- Variables with high variance can dominate the prediction of certain mechanisms of action, showing that the sensitivity of the system depends on the interaction between biological variables and experimental metadata.

Taken together, these relationships show that the system is neither linear nor independent: each element is connected, and any change in inputs or processes can significantly alter predictions. Understanding this dynamic allows us to identify critical points for improvement, such as the sensitivity analysis of `cp_type`, which illustrates how the distinction between control and treatment can modulate the entire flow of information and, consequently, the performance of the model.

3.3 Dependencies and data flows

The MoA Prediction competition workflow can be understood as a system of closely interconnected modules designed to predict the 207 mechanisms of action (MoAs) of compounds on cells.

It all starts with data input, consisting of three main types:

- Gene expression data (g-variables): more than 800 variables describing the expression levels of specific genes in each sample.
- Cellular characteristics (c-variables): approximately 100 variables that reflect the cellular response to treatments.
- Metadatos experimentales: incluyen el tipo de compuesto (`cp_type`), la duración del tratamiento (`cp_time`) y la dosis aplicada (`cp_dose`).

At the same time, the test file maintains the same variable structure, but the MoA labels are not available, so the model must generate predictions for these samples.

With the available data, preprocessing is performed, a critical stage that includes dimensionality reduction, normalization, and experimental variability control. This step allows us to identify important dependencies, such as:

- Correlations between closely related genes.
- Interactions between specific doses and cellular responses.
- Modulating effects of `cp_type` and `cp_time` on gene expression and cellular characteristics.

These dependencies show that small changes in certain variables can significantly alter the prediction of multiple MoAs simultaneously, highlighting the nonlinear and sensitive nature of the system.

Modeling is performed using multi-label supervised learning algorithms, such as neural networks, ensemble models, and One-vs-Rest logistic regression variants. Each output (MoA) depends on a complex set of interrelated inputs, and the presence of correlations between variables means that the prediction of certain MoAs is indirectly influenced by other variables.

Finally, the system moves on to evaluation, generating an activation probability for each combination of unique identifier (`sig_id`) and MoA. These probabilistic predictions are compared with the known truth using the log loss metric, which penalizes both classification errors and inadequate confidence in predictions. The final result is consolidated in a `submission.csv` file, which compiles all predictions for each sample and each MoA.

Taken together, this flow shows how each module depends on the previous one: from input to output, the accuracy of the 207 MoA predictions depends directly on the quality of the preprocessing and the correct capture of the interactions between g-variables, c-variables, and metadata. This highlights that the MoA Prediction competition is a highly interconnected, multi-label, and non-linear system, where understanding critical dependencies is key to optimizing predictions.

3.4 System boundaries and systems engineering perspective

The Mechanism of Action (MoA) Prediction system is conceived as a multi-label probabilistic classifier, designed to estimate the membership of each compound to one or more of the 207 known MoAs, based on tabular experimental data.

Scope and limitations

- The system only predicts MoAs present in the training data, so it does not discover new mechanisms, establish causal relationships, or provide clinical recommendations.
- It only processes the data types for which it was designed: *g-variables* (gene expression), *c-variables* (cellular characteristics), and experimental metadata (`cp_type`, `cp_time`, `cp_dose`). Data outside this domain or inconsistent formats may degrade model performance.
- Data quality and coverage determine reliability: underrepresented MoAs lead to unstable predictions, while noise, imbalance, or batch effects can skew the results.

Actors

Although it is a Kaggle challenge and not a production system, from a systems engineering perspective, the following can be considered conceptual actors:

- **Data scientists:** They manage and process data, train multi-label models, and select features.
- **Analysts and biologists:** interpret the probabilities and patterns detected by the model.

Functionality

- **Input and preprocessing:** tabular data ingestion, normalization, dimensionality reduction, and control of correlations between variables. This stage ensures that the inputs are ready for multi-label modeling.
- **Modeling:** Supervised algorithms (neural networks, ensemble models, multi-label logistic regression) generate probabilities for each MoA, capturing nonlinear interactions among *g-variables*, *c-variables*, and meta-data.
- **Assessment:** Here's the English translation of your text: The generated probabilities are compared with the true labels using *log loss*, a metric that reflects the model's accuracy and confidence. The final output is a `submission.csv` file containing the predictions for all samples and the 207 MoAs.

Dependencies and sensitivity

- Each module depends on the previous one: errors in preprocessing or variable selection directly affect MoA prediction.
- Variables such as `cp_type`, `cp_time`, or dose have modulatory effects that can amplify or attenuate the prediction of certain MoAs.
- The multi-label nature of the problem implies that the predictions of some MoAs are correlated with others, increasing model complexity and sensitivity to the inputs.

Non-functional requirements

- Although it is not deployed as a production system, the conceptual principles are maintained: reproducibility of results, version control for data and models, and documentation of the training *pipeline*.
- Model evaluation relies on standardized metrics and protocols (*log loss*), so the results are relative to the validation methodology used.

4 Complexity, sensitivity, and chaos

4.1 Input Parameter Sensitivity

The Mechanism of Action (MoA) prediction system receives as input a set of parameters that describe both the experimental conditions of each sample and the biological responses measured in the laboratory. Among these parameters, some play a more prominent role in the model’s dynamics and help explain how predictions are generated. The most important include `cp_type`, `cp_time`, `cp_dose`, and the biological variables represented by the `g-` and `c-variables`.

Table 1: Description of the MoA system input parameters

Type	Columns	Description
Categorical	<code>cp_type</code>	Sample type: control or treated with an experimental compound
Categorical	<code>cp_time</code>	Duration of compound exposure (24, 48, 72 hours)
Categorical	<code>cp_dose</code>	Applied dose level (low or high)
Numerical	<code>g-0 ... g-771</code>	Gene expression levels for 772 genes
Numerical	<code>c-0 ... c-99</code>	Cell viability measurements under 100 experimental conditions

Within the system, `cp_type` serves as a filter: it identifies whether the sample corresponds to an experimental treatment (`trt_cp`) or to a control (`ctl_vehicle`). In the competition, only treated samples have MoA labels and are included in the modeling, while control samples function as a reference. Conceptually, when moving from control to treated, MoA prediction is enabled, whereas control samples generate a null output. This demonstrates a sensitivity in terms of prediction activation, although `cp_type` by itself does not determine which MoAs are assigned.

The treatment duration (`cp_time`) directly influences gene expression and cell viability. Different exposure times can modify the activation probability of certain MoAs, showing moderate-to-high sensitivity, especially when interacting with the applied dose and the biological profiles of each sample.

Similarly, `cp_dose` regulates the magnitude of the biological response. Higher or lower doses increase or decrease the probability of certain MoAs, reflecting high sensitivity, since small variations can significantly alter the model’s output.

Finally, the numerical biological variables (`g-` and `c-variables`) represent gene expression and cell viability, and they are the parameters exerting

the greatest influence on predictions. Any change in these values can directly and complexly alter the probabilities of the MoAs, showing the highest sensitivity within the system.

This analysis helps explain how each parameter contributes to the system’s dynamics: `cp_type` regulates which samples generate predictions, `cp_time` and `cp_dose` modulate the intensity of the response, and the `g-` and `c-variables` determine the specific probability patterns for each MoA. The conceptual sensitivity assessment reveals which parameters have the greatest impact on predictions and how they interact with each other and with the biological variables, keeping the explanation fully aligned with the structure and objectives of the MoA competition.

4.2 Random or nonlinear processes

Within the framework of the complexity of the mechanism of action (MoA) prediction problem, random and nonlinear processes are central elements that affect both the algorithms and the biological data. In the machine learning models used in the competition, such as neural networks and ensemble methods, there are stochastic components: random parameter initialization and random selection of training subsets can generate significant variations in the results, even when the input data are identical.

At the same time, biological data present inherently nonlinear relationships. The interaction between gene expression (`g-variables`), cell viability (`c-variables`), and experimental metadata (`cp_type`, `cp_time`, `cp_dose`) does not follow a simple proportional pattern: minimal changes in the activation of a gene or in the applied dose can be unexpectedly amplified, affecting multiple MoAs in an unpredictable way.

The combination of randomness in the algorithms and nonlinearity in the data turns the MoA prediction system into a highly dynamic environment, where uncertainty and complexity are intertwined. For this reason, the competition requires robust and carefully tuned models, capable of capturing high-order interactions and generalizing reliably in the face of variations in data and experimental conditions. This highlights that understanding and managing stochasticity and nonlinear relationships is crucial to achieving accurate and reliable predictions within the context of the competition.

4.3 Feedback, loops, and aspects of chaos

In the MoA competition, feedback is not just a theoretical concept, but a central mechanism that guides the improvement of the predictive system. Technical feedback is reflected in how performance metrics, primarily *log*

loss in cross-validation, allow participants to adjust internal processes. For example, if the model shows poor performance on underrepresented MoAs, participants adjust data preprocessing, such as normalization of columns **g-0** to **g-771** or feature selection, and retrain the model to improve generalization.

Feedback on data quality is also critical. During training and validation, issues such as outliers, batch effects, or class imbalance are detected. These findings feed back into the cleaning steps, such as correction of *batch effects* and filtering of *outliers*, ensuring that the inputs are consistent and that the models learn real patterns instead of noise.

These processes occur in iterative loops, where each cycle contributes to improving prediction:

- **Modeling loop:** the models (neural networks, *gradient boosting*, or ensembles) are repeatedly trained, validated, and tuned on the 875 input parameters (**cp_type**, **cp_time**, **cp_dose**, **g-** and **c-** variables) until the *log loss* stabilizes.
- **Data processing loop:** normalization, dimensionality reduction, and feature selection are applied in successive cycles to optimize the representation of information.
- **Ensemble loop:** multiple trained models are combined, adjusting weights and revalidating the results, with the aim of obtaining a robust ensemble that improves the prediction of both rare and frequent MoAs.

The system also exhibits aspects of chaos, which reflect its intrinsic sensitivity:

- **Sensitivity to splits:** small variations in the train/validation partition can significantly affect the *log loss*, especially for MoAs with few samples.
- **Dependence on drug grouping:** if the same compound appears in both training and validation, performance may be overestimated. Therefore, *drug-aware splitting* is applied to ensure that the same compound is not mixed across partitions, revealing the true generalization capacity.
- **Model stochasticity:** random initializations in neural networks or the processing order in boosting methods generate variations across runs, making it necessary to average or combine predictions.

- **Nonlinear interactions:** the complex relationships between gene expression and cell viability produce unexpected effects; a change in a single gene or in the dose (`cp_dose`) can disproportionately modify the probability of multiple MoAs.

Taken together, these feedbacks, loops, and chaotic dynamics show that the MoA system is dynamic and sensitive, and that improvement depends on carefully planned iterative adjustments to the data and model, always guided by the competition’s official evaluation metrics.

5 Visual representation of the system

5.1 Data flow diagram

Level 0 shows the general interaction of the Mechanism of Action (MoA) Prediction System with its environment. At the center is the system, whose function is to process incoming information and return useful results to external actors.

The external entities that interact with the system are:

- **Users:** Enter experimental data or analysis requests and receive reports with mechanism of action predictions as output.
- **Biological data sources:** Provide experimental information and biological data that are fundamental for the functioning of the system.
- **Knowledge repository:** Contains reference information and annotations of already known mechanisms, which are used to train and validate the system’s models.

Through data flows, the system receives information from these entities, processes it internally, and generates predictions and reports that are delivered to users. This level highlights the system’s relationship with its environment, clearly showing how external inputs are transformed into useful outputs.

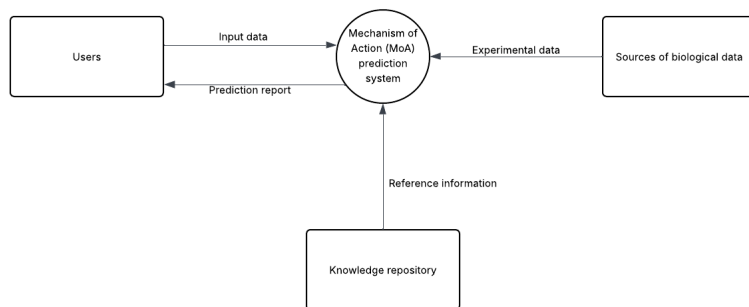


Figure 1: Data flow diagram

Level 1 delves into the internal processes of the system and the flow of data between its components:

- **Data input:** The system receives experimental data from users and biological sources. These include gene expression, cell viability, and metadata about treatment type, dose, and exposure time.
- **Preprocessing:** The input data undergo cleaning, normalization, and dimensionality reduction to eliminate inconsistencies and prepare the information for analysis. This ensures that the machine learning models work with reliable and comparable data.
- **Model training:** Machine learning algorithms are applied using both the preprocessed data and the knowledge repository. The trained models are stored internally, allowing future predictions to be performed without retraining from scratch.
- **Model ensemble and prediction:** Individual models are combined into an ensemble to generate more robust predictions and reduce bias. Each prediction indicates the probability that a specific sample activates one or more mechanisms of action.
- **Evaluation and feedback:** The predictions are compared with known data to adjust the models and improve the overall performance of the system. This feedback cycle ensures that the system learns from previous iterations and maintains its reliability.

- **Output of results:** Finally, the system provides a clear and reliable report to users, representing the predictions as probability vectors for each sample and each mechanism of action.

This workflow ensures that the system meets the objectives of the MoA Prediction competition, reliably and efficiently processing large volumes of data and delivering useful predictions for compound prioritization and biomedical trial design.

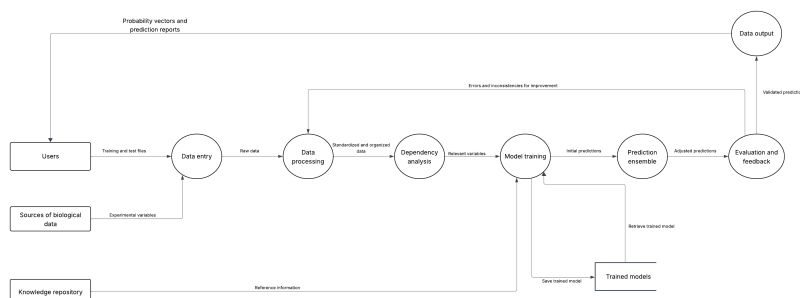


Figure 2: Data flow diagram

5.2 Map of relationships between elements

The representation of the system can be understood as a relationship map where each element depends on the others and contributes to the final result. At the base are the inputs, consisting of gene expression data, cell viability, and experimental metadata (`cp_type`, `cp_time`, `cp_dose`). These inputs do not act in isolation but rather interact with each other and with subsequent processes.

The next element is preprocessing, where the data are normalized, their dimensionality is reduced, and experimental variability is corrected. This stage establishes the conditions for machine learning models to identify relevant patterns and complex relationships among the variables. Any modification or error in this phase directly affects the quality of the predictions.

Next, the processed data enter the modeling module, in which neural networks, gradient boosting, and ensembles capture nonlinear interactions among the inputs and generate probabilistic predictions about the mechanisms of action (MoAs). These predictions are evaluated using metrics such as log loss, forming a feedback cycle that allows adjustment of both preprocessing and model parameters.

Finally, the results are consolidated in the system output, represented by the file `submission.csv`, which organizes the predictions for each sample and

each MoA. This workflow shows that each element depends on the previous one and affects the next, and that the interaction among inputs, processes, and evaluation is fundamental to the system’s performance. Understanding these relationships makes it possible to identify critical points for improvement and ensure that the system operates coherently with the objectives of the MoA competition.

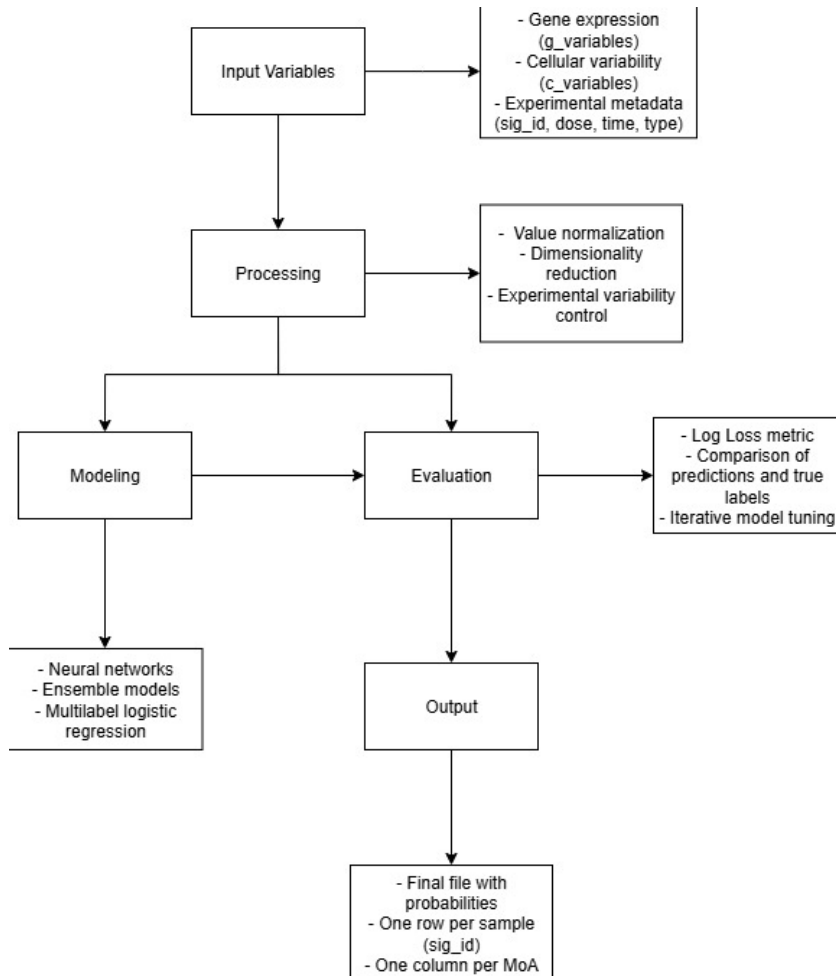


Figure 3: Map of relationships between elements

5.3 Interaction diagram

The diagram shows how different types of data—such as gene expression, cell viability, and experimental conditions—are integrated to relate them to each other and to the possible mechanisms of action of the compounds. These interactions make it possible to identify what effects a treatment may

have and how different mechanisms are connected to each other, ultimately generating a prediction that is compared with known information to validate its accuracy.

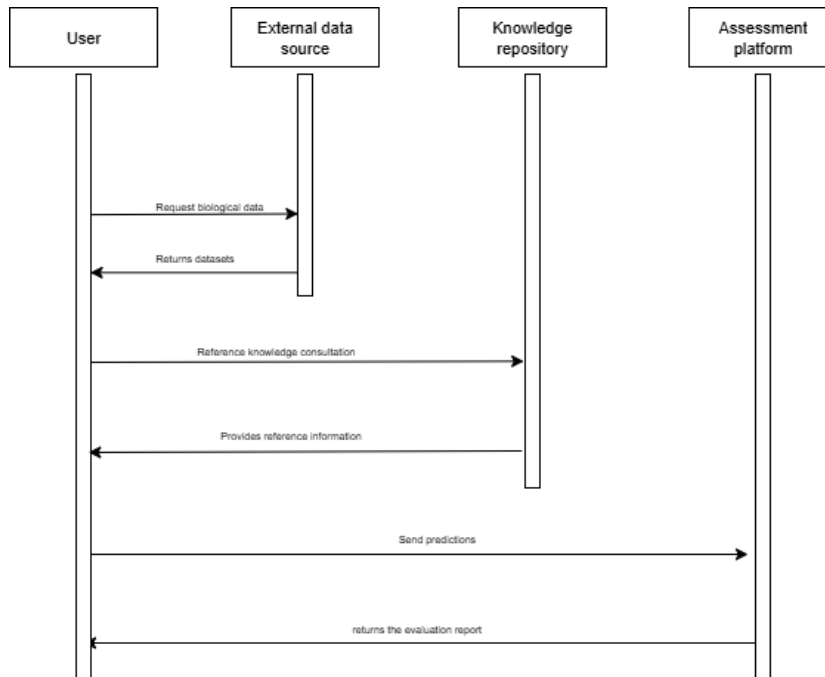


Figure 4: Interaction Diagram

6 Conclusions

- The analysis carried out showed that the MoA Prediction competition represents a complex challenge, where the prediction of mechanisms of action depends on the integration of high-dimensional data with sophisticated modeling processes. It was identified that the system architecture—from the inputs, through preprocessing and modeling, to the outputs—reflects the need for a coherent workflow, where each component conditions the quality of the final predictions. In addition, the importance of critical parameters such as treatment type was highlighted, whose correct interpretation is essential to ensure that control and treated samples are processed in a differentiated manner, avoiding biases that could compromise the validity of the model. This perspective made it possible to understand how the system structure reflects the objectives of the competition and how coordination among modules

directly impacts the reliability of the predictions.

- The analysis of the `cp_type` parameter revealed its central role in the system’s dynamics, functioning as a key determinant that defines whether a sample can generate predictions of mechanisms of action. Its influence not only conditions the activation of MoAs but also modulates the model’s sensitivity to factors such as dose, exposure time, and gene expression. This conclusion highlights that, within a high-dimensional and class-imbalanced environment, understanding the interaction of input parameters is fundamental for properly interpreting the probabilities generated by the model. Thus, the competition requires participants to consider data sensitivity from the perspective of its functional impact on the output, reinforcing the need for preprocessing and modeling strategies adapted to these particularities.
- The project demonstrated that rigorous data handling is the foundation for obtaining reliable predictions in multi-label bioinformatics problems. The distinction between controls and effective treatments, reflected in variables such as `cp_type`, allows for the correct interpretation of cellular responses and gene activation patterns. The organization of the system into inputs, processes, and outputs facilitated the visualization of how each phase contributes to the final result, evidencing the complexity inherent to machine learning applied to pharmacological compounds. Moreover, it was observed that the correct integration of experimental data, robust algorithms, and probabilistic metrics not only enhances model accuracy but also provides a reproducible, scalable analysis framework aligned with the objectives of the competition, reinforcing its applicability in real-world drug discovery scenarios.
- The Mechanisms of Action (MoA) prediction system allows biological and experimental data to be integrated in order to anticipate how compounds act in the body. Its main value lies in discovering relationships between genes, cellular responses, and treatment conditions, which makes it easier to identify the possible mechanisms involved and validate these findings against known information. Overall, it is a tool that provides a clearer and more organized view of the effects of a compound, supporting research and development in biomedicine and pharmacology.

7 References

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